



渣吊金属结构计算书

Metal Structure Calculation Sheet of Slag Crane

1. 起重机的设计要求

1. Crane Design Requirements

1.1 基本参数

1.1 Basic Parameters

起重量: $Q=10t$ 跨度 $L=6.8m$ 起升高度 $H=10m$

Rated Hoisting Load: $Q=10t$, Span: $L=6.8m$, Hoisting Height: $H=10m$

起升速度 $v_q=20m/min$, 起、制动时间 $3s$, 加速度 $a_q=0.11m/s^2$

Hoisting Speed: $v_q=20m/min$, Starting and Braking Time: $3s$, Acceleration: $a_q=0.11m/s^2$

大车运行速度 $v_y=50m/min$, 起、制动时间 $4s$, 加速度 $a_y=0.21m/s^2$

Trolley speed: $v_y=50m/min$, starting and braking time: $4s$, acceleration: $a_y=0.21m/s^2$

小车运行速度 $v_{yt}=25m/min$, 起、制动时间 $4s$, 加速度 $a_y=0.10m/s^2$

Hoist trolley speed: $v_{yt}=25m/min$, starting and braking time: $4s$, acceleration: $a_y=0.10m/s^2$

小车在桥架上的极限位置距离桥架运行轨道中面的距离为 $L_{min}=1.150m$ (左极限), 大车基距 $B=3.2m$, 小车基距 $b=1.8m$ 。

The distance between the limit position of the hoist trolley on the cable tray and the middle surface of the travelling trolley rail of the cable tray is $L_{min}=1.150m$ (left limit), the base of trolley is $B=3.2m$, and the base of hoist trolley is $b=1.8m$.

整机工作级别: A8

Working class of crane: A8

起升机构级别: M8

Hoisting mechanism class: M8

起升机构的起升状态级别为 HC2, 起升驱动装置可作稳定低速运转。

The hoisting status level of the hoisting mechanism is HC2, and the hoisting driving device can be operated stably at low speed.

大车运行机构级别: M8

Trolley travelling mechanism class: M8

小车运行机构级别: M8

Hoist trolley travelling mechanism class: M8

金属结构级别: M8

Metal structure class: M8

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1.2 结构计算遵循的规范

1.2 Specifications Followed by Structural Calculation

EN 13001-1: 2015 Cranes-General design - General principles and requirements

EN 13001-2: 2021 Crane safety - General Design-Load actions

DIN 15080-1: Cranes Steel structures Verification and analyses

1.3 结构设计的许用值

1.3 Allowable Value of Structural Design

1.3.1 材料的许用应力

1.3.1 Allowable Value of Material

起重机结构件的许用应力按式 (1-1) 计算, 并列于表 1.1。

The allowable stresses of crane structural parts are calculated according to Formula (1-1) and are listed in Table 1.1.

$$\sigma_i \leq [\sigma]_i = \frac{R}{\gamma_f \gamma_n} \quad (1-1)$$

式中: R ——强度的标准值。对于屈强比 <0.7 的钢材, $R = \sigma_s$; 对于屈强比 ≥ 0.7 的高强度钢材, $R = 0.5\sigma_s + 0.35\sigma_b$ 。

In the formula: R - standard value of strength. For steels with the yield strength to tensile strength ratio <0.7 , $R = \sigma_s$; for high-strength steels with the yield strength to tensile strength ratio $\geq$ of 0.7, $R = 0.5\sigma_s + 0.35\sigma_b$.

σ_s , σ_b ——钢材屈服强度和抗拉强度, MPa;

σ_s , σ_b - Yield strength and tensile strength of steel, MPa;

γ_f ——强度系数, 按表 1.1 选择;

γ_f - Strength coefficient, selected according to Table 1.1;

γ_n ——高危系数, 取 $\gamma_n = 1.1$ 。

γ_n - High-risk coefficient, $\gamma_n = 1.1$.

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表 1.1 强度系数
Table 1.1 Strength Coefficient

工况 Operating Condition	A	B	C
强度系数 Strength Coefficient	1.48	1.34	1.22

表 1.2 各工况下的许用应力
Table 1.2 Allowable Stress under Each Working Condition

材料名称 Material Name	板厚 Plate Thickness mm	屈服强度 Yield Strength MPa	抗拉强度 Tensile Strength MPa	屈服比 Yield Ratio	许用应力 $[\sigma]$, MPa Allowable Stress $[\sigma]$, MPa		
					工况 A Working Condition A	工况 B Working Condition B	工况 C Working Condition C
S355JR	≤16	355	470~630	≤0.7	218	240	264
	>16~40	345			211	234	257
	>40~63	335			205	227	249

1.3.2 设计的许用刚度 1.3.2 Allowable Design Strength

结构的静刚度以结构在规定载荷作用下指定截面的静态挠度倒数 $1/Y_L$ 来表征。静刚度的设计准则为：

The static rigidity of the structure is characterized by the reciprocal of the static deflection $1/Y_L$ of the specified section under the specified load. The design criteria of static rigidity are:

$$Y_L \leq [Y_L] = 6900/1000 = 6.8\text{mm}$$

式中： Y_L ——垂直静挠度； $[Y_L]$ ——许用挠度值，跨中按 $S/1000$ 计，其中 S 为跨度。

In the formula: Y_L - vertical static deflection; $[Y_L]$ - allowable deflection value, the mid-span shall be calculated according to $S/1000$, wherein S is the span.

对于桥式类型起重机，当满载小车位于跨中有效工作位置时，主梁的对应截面在移动载荷（额定起升载荷和小车自重载荷之和，不计动载系数）作用下的挠度值。

For the bridge crane, when the hoist trolley at full load is located at the effective working position in the span, the corresponding section of the main beam is deflected under the action of moving load (the sum of the rated hoisting load and the self-weight load of the hoist trolley, excluding the dynamic load coefficient).

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2. 载荷组合和计算载荷

2. Load Combination and Calculation Load

2.1 载荷组合

2.1 Load Combination

2.1.1 载荷工况描述

2.1.1 Description of Loaded Working Condition

表 2.1 结构计算载荷工况
Table 2.1 Structural Calculation Load Conditions

载荷工况 Load Conditions	起重机作业状态描述 Description of Crane Operation Status
A1	大车不动，小车运行至规定位置（跨中或端部极限位置）起升或下降载荷 The trolley is not operated, and the hoist trolley is operated to the specified position (mid-span or end limit position) for hoisting or lowering load.
A2	大车运行制动，小车位于规定位置起升或下降载荷 The trolley is in travelling braking, and the hoist trolley is located at the specified position for hoisting or lowering load.
A3	小车位于跨中满载起升或下降制动，同时大车平稳制动 The hoist trolley is in the span for full load hoisting or lowering braking, while the trolley is in stable braking.
B1	小车位于跨端极限位置满载下降制动，同时大车稳定运行 The hoist trolley is at the limit position of the span end, and the trolley is in stable travelling.
B2	小车位于跨端极限位置满载下降制动，同时大车偏斜运行 The hoist trolley is at the limit position of the span end, and the trolley is in deflection travelling.
C1	小车位于跨中，平稳起吊静态试验载荷，小车和大车平稳制动 The hoist trolley is located in the span, stable hoisting and static test load, and the hoist trolley and trolley are in stable braking.
C2	小车位于跨中，起吊动态试验载荷起升或下降制动，同时大车运行制动 The hoist trolley is located in the span for hoisting or lowering braking at the dynamic test load during the hoisting, while the trolley is operated for braking.

2.1.2 计算载荷组合

2.1.2 Calculation of Load Combination

桥式起重机的桥架结构主要由主梁和端梁构成。计算的载荷组合见表 2.2。

The bridge structure of bridge crane is mainly composed of main beam and end beam. The calculation load combination is as shown in Table 2.2.

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表 2.2 计算载荷组合
Table 2.2 Calculation Load Combination

	A 类 Class A			B 类 Class B		C 类 Class C		
	A1	A2	A3	B1	B2	C1	C2	
桥架自重载荷 DL Self-weight load of cable tray DL	ϕ_1	ϕ_1	ϕ_4	ϕ_1	ϕ_4	ϕ_1	ϕ_4	
移动载荷 TL Moving load TL	ϕ_1	ϕ_1	ϕ_4	ϕ_1	ϕ_4	ϕ_1	ϕ_4	
起升载荷 LL+LS Hoisting load LL+LS	ϕ_2	ϕ_2	ϕ_2	ϕ_2	ϕ_2	1.25	1.1 ϕ_6	
小车运行惯性载荷 LATT Inertial load of hoist trolley travelling LATT	ϕ_5			ϕ_5		1		
大车运行惯性载荷 LATG Inertial load of trolley travelling LATG		1	ϕ_5	ϕ_5	ϕ_5	1	ϕ_5	
大车运行偏斜载荷 SKW Deflection load of trolley travelling SKW					1		1	

2.2 计算载荷

2.2 Calculation Load

2.2.1 动载系数

2.2.1 Dynamic Load Coefficient

(1) 起升冲击系数 ϕ_1 :

(1) Hoisting Impact Coefficient ϕ_1 :

当物品离地起升时，或将悬吊在空中的部分物品突然卸除时，或悬吊在空中的物品下降制动时，起重机本身（主要是对金属结构）的自重因出现振动而产生脉冲式增大或减小的动力响应。取 $\phi_1=1.1$ 。

When the object is hoisted from the ground, when the part thereof suspended in the air is suddenly removed, or when the object suspended in the air is lowered, the weight of the crane itself (mainly on the metal structure) will produce the dynamic response of pulse increase or decrease due to the vibration. $\phi_1=1.1$.

(2) 起升冲击载荷系数 ϕ_2 :

(2) Hoisting Impact Load Coefficient ϕ_2 :

当物品无约束地起升离地时，物品的惯性力会使起升载荷出现动载增长的作用。按起升状态级别 HC2，则有

When the hoisting object is hoisted off the ground without restraint, the inertia force of the object will cause the hoisting load to increase in the dynamic load. If the hoisting status level is HC2,

$$\phi_2 = \phi_{2\min} + \beta_2 v_q = 1.1 + 0.34 \times 20/60 = 1.213$$

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式中: $\phi_{2\min}$ ——与起升状态级别相对应的起升动载系数的最小值, 取为 1.1。

In the formula: $\phi_{2\min}$ —— the minimum value of the hoisting dynamic load coefficient of corresponding to the hoisting status level is 1.1.

β_2 ——按起升状态级别设定的系数, 取为 0.34;

β_2 - Coefficient set according to the level of the hoisting status: 0.34;

v_q ——稳定起升速度, m/s, 与起升机构驱动控制型式及操作方法有关。

v_q ——The stable hoisting speed (m/s) is related to the driving control type and operation method of the hoisting mechanism.

(3) 运行冲击系数 ϕ_4 :

(3) Travelling Impact Coefficient ϕ_4 :

起重机或起重机的部分装置, 在不平的道路或轨道上运行时所发生的垂直冲击动力效应, 见图 2.1。

The vertical impact dynamic effect of crane or partial device of the crane during the travelling on the uneven road or trolley rail is shown in Fig. 2.1.

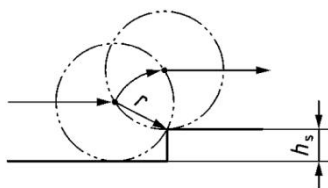


图 2.1 运行中高低差

Fig. 2.1 Height-low Difference in Travelling

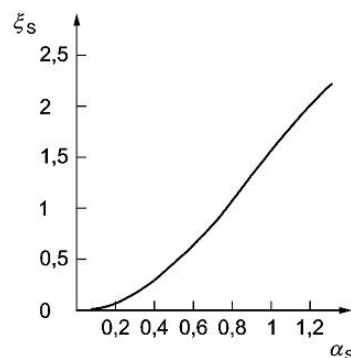


图 2.2 曲线系数

Fig. 2.2 Curve Coefficient

根据 BS EN 13001-2 中 4.2.2.4 条款计算如下。

The calculation is as follows in accordance with 4.2.2.4 of BS EN 13001-2.

频率 $f_q=10\text{Hz}$, 轨道高低差 $h_s=1\text{mm}$, 大车车轮半径 $r_y=125\text{mm}$, 小车车轮半径 $r_{yt}=100\text{mm}$,

则由 $\alpha_s = \frac{2f_q h_s}{v} \sqrt{\frac{2r}{h_s}}$ 和图 2.2 得到 ξ_s 。

If frequency: $f_q=10\text{Hz}$, trolley rail height-low difference: $h_s=1\text{mm}$, wheel radius of trolley:

$r_y=125\text{mm}$, wheel radius of hoist trolley: $r_{yt}=100\text{mm}$, $\alpha_s = \frac{2f_q h_s}{v} \sqrt{\frac{2r}{h_s}}$ will be obtained

according to ξ_s and figure 2.2.

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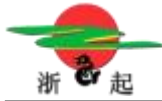
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$$\text{大车运行时 } \phi_4 = 1 + \left(\frac{\pi}{2}\right)^2 \frac{v_y^2}{gr} \xi_s = 1 + \left(\frac{\pi}{2}\right)^2 \frac{(50/60)^2}{9.81 \times 0.125} \times 0.28 = 1.26$$

$$\phi_4 = 1 + \left(\frac{\pi}{2}\right)^2 \frac{v_y^2}{gr} \xi_s = 1 + \left(\frac{\pi}{2}\right)^2 \frac{(50/60)^2}{9.81 \times 0.125} \times 0.28 = 1.26 \text{ during the travelling of trolley}$$

$$\text{小车运行时 } \phi_4 = 1 + \left(\frac{\pi}{2}\right)^2 \frac{(25/60)^2}{9.81 \times 0.1} \times 0.83 = 1.24$$

$$\phi_4 = 1 + \left(\frac{\pi}{2}\right)^2 \frac{(25/60)^2}{9.81 \times 0.1} \times 0.83 = 1.24 \text{ during the travelling of hoist trolley}$$

(5) 动载系数 ϕ_5

(5) Dynamic load coefficient ϕ_5

由驱动机构加速或减速、起重机意外停机或传动机构突然失效等原因在起重机中引起的弹性效应，取为 1.5。

The elastic effect in the crane caused by the acceleration or deceleration of the driving mechanism, the unplanned shutdown of the crane or the sudden failure of the transmission mechanism is 1.5.

2.2.2 桥架自重载荷 DL

2.2.2 Self-weight Load of Cable Tray DL

主要有：桥架自重（包括有主梁、轨道、走台、栏杆、电气及导电装置等）、大车运行机构、操纵司机室以及电气室等重量。

The self-weight of the cable tray (including the main beam, trolley rail, platform, tracks, electrical and conductive devices, etc.), and the weight of the trolley travelling mechanism, the operator cabin and the electrical room are mainly included.

表 2.3 自重载荷
Table 2.3 Self-weight Load

项目 Item	重量 Weight	单位 Unit
桥架 Cable tray	3.5	吨 Ton
电气 Electrical device	1.7	吨 Ton
大车运行机构 Trolley travelling mechanism	1.3	吨 Ton
总和 Sum	6.5	吨 Ton

桥架及其上部件的自重以均布载荷考虑， $q_G = 5.2 \times 9.81 / 6.8 / 2 = 3.75 \text{ kN/m}$ （单根主梁）。大车运行机构的自重作用在桥架的支座处。

The self-weight of crane tray and the components thereof is considered as the uniform distribution load, $q_G = 5.2 \times 9.81 / 6.8 / 2 = 3.75 \text{ kN/m}$ (single main beam). The self-weight of the trolley travelling mechanism acts on the support of cable tray.

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2.2.3 移动载荷 TL+LS+LL
2.2.3 Moving Load TL+LS+LL

主梁上的移动载荷由起升机构、小车的自重和起升载荷组成。起升载荷是由起升质量引起的重力。

The moving load on the main beam is composed of the self-weight of the hoisting mechanism and hoist trolley, and the hoisting load. The hoisting load is the gravity caused by the hoisting mass.

表 2.4 移动载荷
Table 2.4 Moving Load

项目 Item	重量 Weight	单位 Unit
小车自重 TL Self-weight of hoist trolley TL	4.9	吨 Ton
起升载荷 Hoisting load	3.8	吨 Ton
电动抓斗 LS Motor driven grab LS	3.8	吨 Ton
起重量（包括抓斗）LS+LL Rated hoisting load (including grab) LS+LL	10.0	吨 Ton

不考虑动载效应的移动载荷 $\Sigma P = (4.9+10) \times 9.81 = 146.169 \text{ kN}$

Moving load without considering the dynamic load effect: $\Sigma P = (4.9+10) \times 9.81 = 146.169 \text{ kN}$

根据简化假定，小车轮压按四个均等的集中力考虑，则

According to the simplified assumption, the hoist trolley wheel load shall be considered as four uniform concentrated force,

$$P_d = \frac{\phi_2 \cdot (LL + LS) + \phi_4 \cdot TL}{4} = (1.213 \times 10 + 1.26 \times 4.9) \times 9.81 / 4 = 44.89 \text{ kN}$$

2.2.4 运行惯性载荷 LATG/LATT
2.2.4 Travelling Inertial Load LATG/LATT

根据刚体动力学原理进行计算，还需考虑结构的动载效应，即乘以 ϕ_5 加以考虑。按打滑条件，计算公式如下所示：

According to the principle of rigid body dynamics, the dynamic load effect of the structure shall also be considered, that is, it is multiplied by ϕ_5 . According to the slipping conditions, the calculation formula is as follows:

$$LATT = ma_{yt} < \mu P_L, \quad LATG = ma_y < \mu P_L$$

式中： μ ——系数，当运行机构总轮数的 1/2 为驱动轮时，取 $\mu = 0.1$ （室内）；

In the formula: μ - coefficient, when half of the total number of the wheels in the travelling mechanism refers to the driving wheel, $\mu = 0.1$ (indoor);

P_L ——小车车轮上的总垂直静轮压；

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P_L - The total vertical static wheel load on the hoist trolley wheel;

q —— 桥架自重载荷。

q - Self-weight load of cable tray.

$$LATT = 0.1 \times (4.9 + 10) \times 9.81 = 14.62 \text{ kN}$$

$$LATG = 0.1 \times (4.9 + 10 + 6.5) \times 9.81 = 20.99 \text{ kN}$$

小车运行机构起动或制动时，由移动载荷引起的惯性载荷是以集中力 P_H 的方式沿水平方向作用在跨度中或端部，并通过小车轮缘与轨道侧向接触从而传递给主梁；大车运行机构起动或制动时，桥架部分质量引起的水平惯性载荷是以均布载荷 q_H 的方式作用在主梁上。 P_H 和 q_H 值可以表达为

When the hoist trolley travelling mechanism is in the starting or braking status, the inertial load caused by the moving load acts in the span or at the end in the horizontal direction by means of the concentrated force P_H , and is transmitted to the main beam through the lateral contact between the hoist trolley rim and the trolley rail. When the trolley travelling mechanism is in the starting or braking status, the horizontal inertial load caused by part of the mass of the cable tray is applied to the main beam by means of the uniform load q_H . P_H The value q_H can be expressed as follows:

$$P_H = \phi_5 LATT / 4 = 1.5 \times 14.62 / 4 = 5.48 \text{ kN}$$

$$q_H = \phi_5 \frac{LATG}{L} = 1.5 \times 20.99 / 6.8 / 2 = 2.31 \text{ kN/m}$$

当满载小车制动时，它将产生沿主梁轨道方向的水平惯性载荷 P'_H 。按打滑条件，作用在一根主梁上的 P'_H 可按下面公式计算：

During the braking of the hoist trolley at full load, it will produce the horizontal inertial load P'_H along the direction of the main beam trolley rail. According to the slipping condition, P'_H acting on the main beam can be calculated according to the following formula:

$$P_{dy} = \frac{\phi_4 TL + \phi_2 (LS + LL)}{4} = (1.26 \times 4.9 + 1.213 \times 10) \times 9.81 / 4 = 44.89 \text{ kN}$$

$$P'_H = \mu P_{dy} = 0.1 \times 44.89 = 4.489 \text{ kN}$$

式中： P_{dy} ——作用在一根主梁上的小车驱动轮轮压之和；

In the formula: P_{dy} - the sum of the wheel load of the hoist trolley driving wheel acting on the main beam;

ϕ_2, ϕ_4 —— 起升载荷动载系数和运行冲击系数。

ϕ_2, ϕ_4 - Hoisting load and dynamic load coefficient and travelling impact coefficient.

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2.2.5 试验载荷 P_t

2.2.5 Test Load P_t

静载试验载荷取为 1.25 起升载荷，动态试验载荷取为 1.1 起升载荷。在验算时，动态试验载荷应乘以试验载荷起升动载系数 $\phi_6 = (1 + \phi_2) / 2 = 1.11$ 。

The static load test load is the hoisting load of 1.25, and the dynamic test load is the hoisting load of 1.1. In the checking calculation, the dynamic test load shall be multiplied by the test load and hoisting dynamic load coefficient of $\phi_6 = (1 + \phi_2) / 2 = 1.11$.

2.2.6 偏斜运行时的水平侧向力 P_s

2.2.6 Horizontal Lateral Force during the Deflection Travelling P_s

大车偏斜侧向力 P_s 是以一对力偶的形式作用在同一侧端梁的两个车轮上，或作用在桥架的对角线位置的两个轮上，见图 2.2。

The deflection lateral force P_s of trolley is in the form of a pair of couples acting on the two wheels of the end beam on the same side, or on the two wheels of the diagonal position of the cable tray. See Fig. 2.2.

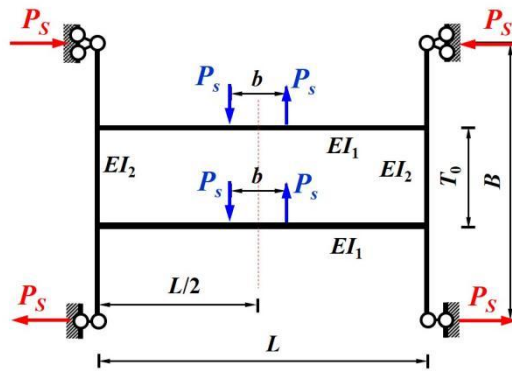


图 2.2 大车偏斜运行时的水平侧向力作用图

Fig. 2.2 Horizontal Lateral Force Action during Trolley Deflection Travelling

根据 BS EN 13001-2 中 4.2.3.4 条款计算如下：

The calculation is as follows in accordance with 4.2.3.4 of BS EN 13001-2:

轨道采用宽度为 $b_h = 70\text{mm}$ 的方钢，起重机单侧轨道上车轮数量 $n = 2$ 。

The trolley rail adopts the square steel with the width of 70mm, and the number of wheels on the single side trolley rail of the crane is $n = 2$.

轮对 i 与导向装置的距离 $d_i = 0.0, 3.2\text{m}$ ，前后导向轮之间的距离 $W_b = 3.2\text{m}$ 。按偏斜角 $\alpha = 0.01\text{rad}$ ，则

The distance between the wheelset i and the guide device is $d_i = 0.0, 3.2\text{m}$, and the distance between the front and rear guide wheels is $W_b = 3.2\text{m}$. If the declination angle is $\alpha = 0.01\text{rad}$,

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系数 $f = \mu_0 \times [1 - e^{(-250\alpha)}] = 0.3 \times (1 - e^{-1.64}) = 0.242$

Coefficient $f = \mu_0 \times [1 - e^{(-250\alpha)}] = 0.3 \times (1 - e^{-1.64}) = 0.242$

在单个车轮上的平均偏斜载荷 $P_s = v \times f \times mg / n = 0.74 \times 0.242 \times (6.8 + 10) \times 9.81 / 2 = 14.65 \text{ kN}$

The average eccentric load on single wheel is $P_s = v \times f \times mg / n = 0.74 \times 0.242 \times (6.8 + 10) \times 9.81 / 2 = 14.65 \text{ kN}$

3. 主梁的计算简图和计算载荷的确定

3. Calculation Diagram of Main Beam and Determination of Calculation Load

3.1 主梁的截面参数

3.1 Sectional Parameters of Main Beam

主梁的截面如图 3.1 所示，坐标系如图 3.2 所示，截面参数如表 3.1。

The section of the main beam is as shown in Fig. 3.1, the coordinate system is as shown in Fig. 3.2, and the sectional parameters are as shown in Table 3.1.

表 3.1 截面参数表
Table 3.1 Sectional Parameters

截面 Section	主截面 Main Section	与端梁连接截面 Section Connected with the End Beam	
梁高, mm Beam height, mm	566	250	
距中性轴最大边缘距离 $y_{\max}$, mm Maximum edge distance from the neutral axis $y_{\max}$, mm	330	330	水平面内 In the horizontal plane
距中性轴最大边缘距离 $z_{\max}$, mm Maximum edge distance from the neutral axis, mm	6	6	铅锤面内 In the plumb plane
翼缘面积 A_y , mm ² Flange area A_y , mm ²	288.256	103.285	
腹板面积 A_z , mm ² Web area A_z , mm ²	176.0518	170.99593	
扇形惯性极 2Ω , mm ² Sector inertia pole 2Ω , mm ²	5.04E+03	6.60E+03	
轴惯性矩 I_y , mm ⁴ Axial moment of inertia I_y , mm ⁴	7.70E+03	3.22E+03	
轴惯性矩 I_z , mm ⁴ Axial moment of inertia, mm ⁴	2.78E+05	1.20E+05	
截面模量 W_{y1} , mm ³ Section modulus W_{y1} , mm ³	5.86E+08	1.07E+08	

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截面模量 W_{y2} , mm^3 Section modulus, mm^3	1.60E+08	1.10E+08	
截面模量 W_{z1} , mm^3 Section modulus, mm^3	2.03E+06	1.04E+06	
截面模量 W_{z2} , mm^3 Section modulus, mm^3	2.11E+06	7.29E+05	

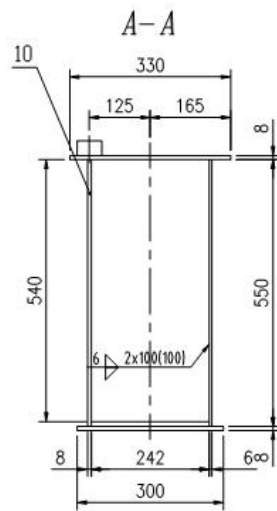


图 3.1 主梁截面
Fig. 3.1 Main Beam Section

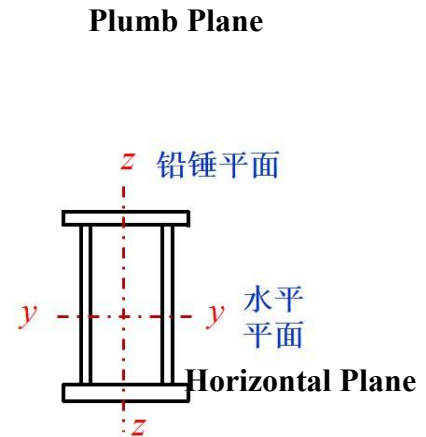


图 3.2 截面坐标系
Fig. 3.2 Sectional Coordinate System

3.2 主梁的计算简图

3.2 Calculation Diagram of Main Beam

主梁强度计算简图如图 3.2 所示，图中给出了桥架在铅锤和水平平面内所承受的载荷，以及小车作用的位置。

The strength calculation diagram of the main beam is as shown in Fig. 3.2, which shows the load borne by the bridge frame in the plumb and horizontal planes, and the action position of the hoist trolley.

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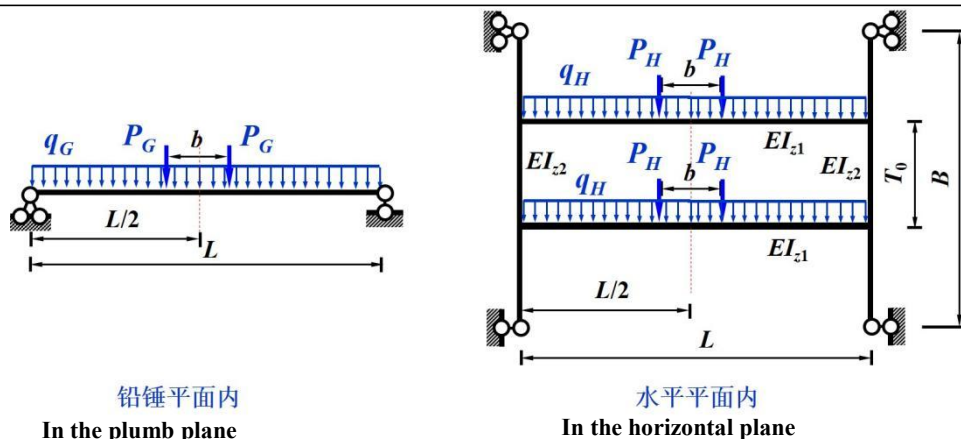


图 3.2 主梁的计算简图

Fig. 3.2 Calculation Diagram of Main Beam

由于本方案采用的是偏轨梁，因此还存在扭矩，偏轨的距离为 125mm。

As the deflection beam is adopted in this scheme, there is still torque, and the deflection distance is 125mm.

3.3 主梁的强度验算原理

3.3 Strength Checking Principle of Main Beam

强度计算的截面为：当小车位于跨中时的 I-I 和 II-II 截面，以及小车位于极限位置时的 III-III 截面。各截面所考虑的

The sections for strength calculations are I-I and II-II sections when the hoist trolley is in span, and III-III section when the hoist trolley is at the limit position. Considered for each section

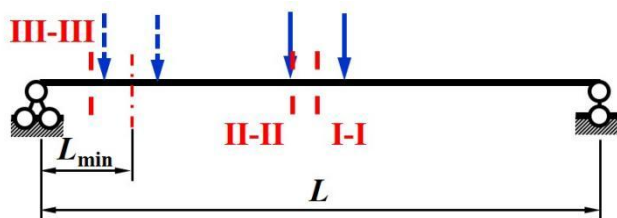


图 3.3 计算截面

Fig. 3.3 Calculation Section

(1) I-I 截面，由于垂直方向和水平方向最大弯矩同时作用，在主梁截面的翼缘板中引起的最大正应力为：

(1) For the section I-I, due to the simultaneous action of the maximum bending moments in the vertical and horizontal directions, the maximum normal stress caused in the flange plate of the main beam section is:

$$\sigma = \frac{\sum M_y}{W_y} + \frac{\sum M_z}{W_z} \leq [\sigma] \quad (3-1)$$

式中： $\sum M_y, \sum M_z$ ——分别为由垂直载荷和水平载荷在跨中引起的最大弯矩；

In the formula: $\sum M_y, \sum M_z$ - the maximum bending moment in span caused by vertical load and

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horizontal load respectively;

(2) II-II 截面, 主梁的截面内同时存在弯矩和剪力的作用, 应验算截面中由弯曲正应力 σ 和剪应力 τ 产生的复合应力, 可按下式计算:

(2) There are both bending moment and shearing force in the section II-II of and the section of the main beam. The combined stress generated by the normal stress σ and shearing stress τ in bending in the section shall be checked, which can be calculated according to the following formula:

$$\sqrt{\sigma^2 + 3\tau^2} \leq [\sigma] \quad (3-2)$$

式中: σ ——由垂直载荷和水平载荷在 II-II 截面危险点处引起的弯矩正应力;

In the formula: σ - the normal stress in bending caused by vertical and horizontal loads at the danger point of section II-II ;

τ ——由垂直载荷在 II-II 截面的危险点处引起的剪应力。

τ - Shearing stress caused by vertical load at the danger point of section II-II .

(3) III-III 截面, 主要验算跨端截面由垂直剪力 Q_y 引起的腹板最大剪应力, 应满足:

(3) In the III-III section, the checking calculation is mainly performed for the maximum shearing stress of the web caused by vertical shearing force Q_y across the end section shall meet the following requirements:

$$\tau = \frac{Q_z S_y(\omega)}{I_y \sum t_w} \leq [\tau] \quad (3-3)$$

式中: Q_z ——由垂直载荷引起的计算截面上的最大剪力;

In the formula: Q_z - maximum shearing force on the calculation section caused by the vertical load;

$S_y(\omega)$ ——计算剪应力处以上截面对水平轴 $y-y$ 的静矩;

$S_y(\omega)$ - Calculate the static moment of the section above the shearing stress with respect to the horizontal axis $y-y$;

$\sum t_w$ ——主梁腹板的总厚度, 即为两块腹板的厚度 $2t_w$;

$\sum t_w$ - Total thickness of the main beam web, just the thickness $2t_w$ of two webs;

$[\tau]$ ——结构设计的控制应力, $[\tau] = [\sigma] / \sqrt{3}$ 。

$[\tau]$ - Control stress of structural design, $[\tau] = [\sigma] / \sqrt{3}$.

3.4 强度计算结果和校核

3.4 Strength Calculation Result and Check

各工况下, 危险截面上的应力情况如列于表 3.2。

The stress conditions on the dangerous section under various working conditions are as shown in Table 3.2.

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表 3.2 应力汇总表, MPa
Table 3.2 Stress Summary, MPa

载荷组合 Load Combination	危险截面 Dangerous Section	最大正应力 Maximum Normal Stress	最大切应力 Maximum Shearing Stress	复合应力 Combined Stress	强度判断 Strength Judgment
A1	截面 I-I Section I-I	90.60	6.44	91.29	[σ]=218 [τ]=126 OK!
	截面 II-II Section II-II	89.79	11.55	91.99	
	截面 III-III Section III-III	35.93	37.23	73.81	
A2	截面 I-I Section I-I	70.19	6.44	71.07	
	截面 II-II Section II-II	69.22	12.97	72.77	
	截面 III-III Section III-III	2.67	42.38	73.46	
A3	截面 I-I Section I-I	76.41	6.72	77.30	
	截面 II-II Section II-II	75.24	13.65	78.87	
	截面 III-III Section III-III	3.65	44.76	77.61	
B1	截面 I-I Section I-I	96.79	6.44	97.43	[σ]=240 [τ]=139 OK!
	截面 II-II Section II-II	95.73	12.98	98.34	
	截面 III-III Section III-III	38.87	42.40	83.10	
B2	截面 I-I Section I-I	77.58	6.72	78.45	
	截面 II-II Section II-II	76.41	13.65	79.98	
	截面 III-III Section III-III	5.16	44.76	77.70	
C1	截面 I-I Section I-I	68.32	6.58	69.26	[σ]=264 [τ]=152 OK!
	截面 II-II Section II-II	67.34	13.22	71.13	
	截面 III-III Section III-III	3.33	43.16	74.83	
C2	截面 I-I Section I-I	78.32	6.74	79.18	
	截面 II-II Section II-II	77.15	13.67	80.70	
	截面 III-III Section III-III	6.15	44.84	77.91	

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3.5 刚度校核

3.5 Rigidity Check

仅考虑移动载荷引起的下挠度。

The down deflection caused by the moving load shall be considered only.

$$\alpha = (L-b)[3L^2 - (L-b)^2] / 2L^3 = \frac{(6.8-1.8)[3 \times 6.8^2 - (6.8-1.8)^2]}{2 \times 6.8^3} = 0.904$$

$$\text{静刚度 } Y_L = \frac{\alpha \Sigma PL^3}{48EI} = \frac{0.904 \times 146.169 \times 10^3 / 2 \times 6.8^3 \times 10^9}{48 \times 2.06 \times 10^5 \times 5.86 \times 10^8} = 3.585 \text{ mm}$$

$$\text{Static rigidity } Y_L = \frac{\alpha \Sigma PL^3}{48EI} = \frac{0.904 \times 146.169 \times 10^3 / 2 \times 6.8^3 \times 10^9}{48 \times 2.06 \times 10^5 \times 5.86 \times 10^8} = 3.585 \text{ mm}$$

4. 端梁的设计和计算

4. Design and Calculation of End Beam

4.1 计算简图和作用载荷

4.1 Calculation Diagram and Action Load

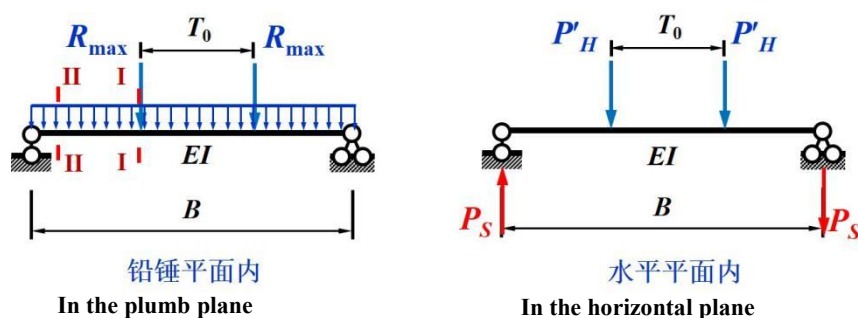


图 4.1 端梁的计算简图

Fig. 4.1 Calculation Diagram of End Beam

端梁的计算简图如图 4.1 所示。作用在端梁上的载荷有自重载荷 q_G 、主梁的支座反力 $R_{\max}$ （小车在极限位置时主梁的最大轮压，见表 4.1）、小车运行的惯性载荷 P'_H 和作用在大车运行机构上的偏斜载荷 P_S 。以上均需考虑动载效应。

The calculation diagram of end beam is as shown in Fig. 4.1. The load acting on the end beam includes the self-weight load q_G , reaction at support of main beam $R_{\max}$ (maximum wheel load of the main beam when the hoist trolley is at the limit position, see Table 4.1), inertial load P'_H of the hoist trolley during the travelling and deflection load P_S acting on the travelling mechanism of the crane. The dynamic load effect shall be considered for all the above.

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表 4.1 各工况下的支座反力 $R_{\max}$, kN

Table 4.1 Reaction at Support under Each Working Condition $R_{\max}$, kN

载荷工况 Load Conditions	A1	A2	A3	B1	B2	C1	C2
支座反力 Reaction at Support	89.16	89.16	94.76	89.16	94.76	90.66	94.923

4.2 端梁的截面参数

4.2 Sectional Parameters of End Beam

端梁的截面为矩形钢 300×200×8，其截面参数见表 4.2。

The section of end beam is the rectangular steel ×200×8, and the sectional parameters are shown in Table 4.2.

表 4.2 截面参数表

Table 4.2 Sectional Parameters

截面 Section	主截面 Main Section	与端梁连接截面 Section Connected with the End Beam	
梁高, mm Beam height, mm	300	112.5	
距中性轴最大边缘距离 $y_{\max}$, mm Maximum edge distance from the neutral axis , mm	200	200	水平面内 In the horizontal plane
距中性轴最大边缘距离 $z_{\max}$, mm Maximum edge distance from the neutral axis , mm	8	8	铅锤面内 In the plumb plane
翼缘面积 A_y , mm ² Flange area , mm ²	150	39.55	
腹板面积 A_z , mm ² Web area , mm ²	100	100	
扇形惯性极 2Ω , mm ² Sector inertia pole , mm ²	4.80E+03	1.60E+03	
轴惯性矩 I_y , mm ⁴ Axial moment of inertia , mm ⁴	3.20E+03	2.25E+03	
轴惯性矩 I_z , mm ⁴ Axial moment of inertia , mm ⁴	1.12E+05	6.67E+03	
截面模量 W_{y1} , mm ³ Section modulus , mm ³	9.88E+07	4.78E+06	
截面模量 W_{y2} , mm ³ Section modulus , mm ³	5.26E+07	2.42E+07	
截面模量 W_{z1} , mm ³ Section modulus , mm ³	6.59E+05	1.21E+05	

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截面模量 W_{z2} , mm ³ Section modulus , mm ³	6.59E+05	6.55E+04	
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4.4 强度校核

4.4 Strength Check

端梁强度计算部位为：① 弯矩最大且安装拼接截面 $I-I$ （主梁与端梁的连接处）；② 支承截面 $II-II$ ，见图 4.1。

The calculation position of the end beam strength is: ① Maximum bending moment and installed splicing section $I-I$: (connection between the main beam and the end beam); ② supporting section: $II-II$. See Fig. 4.1.

各工况下，危险截面上的应力情况如表 4.3。

The stress conditions on the dangerous section under various working conditions are as shown in Table 4.3.

表 4.3 应力汇总表, MPa
Table 4.3 Stress Summary, MPa

载荷组合 Load Combination	危险截面 Dangerous Section	最大正应力 Maximum Normal Stress	最大切应力 Maximum Shearing Stress	合成应力 Resulting Stress	强度判断 Strength Judgment
A1	截面 I-I Section I-I	93.90	32.57	109.54	[σ]=218 [τ]=126 OK!
	截面 II-II Section II-II	0.00	112.52	112.52	
A2	截面 I-I Section I-I	58.20	28.26	76.05	
	截面 II-II Section II-II	2.56	40.19	69.66	
A3	截面 I-I Section I-I	65.00	30.06	83.29	
	截面 II-II Section II-II	3.84	42.76	74.15	
B1	截面 I-I Section I-I	63.93	32.57	85.27	[σ]=240 [τ]=139 OK!
	截面 II-II Section II-II	0.00	112.52	112.52	
B2	截面 I-I Section I-I	81.92	30.22	97.22	
	截面 II-II Section II-II	3.84	42.87	74.35	
C1	截面 I-I Section I-I	60.32	33.04	83.15	[σ]=264 [τ]=152 OK!
	截面 II-II Section II-II	2.56	113.19	196.06	
C2	截面 I-I Section I-I	87.12	34.65	105.79	
	截面 II-II Section II-II	3.84	115.23	199.61	

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5. 主梁和端梁螺栓连接的校核

5. Connection Check of Main Beam and End Beam

5.1 计算载荷

5.1 Calculation Load

螺栓群受到的内力是剪力（小车、大车运行惯性引起 Q_x 、 Q_y 、 M_z ）、压力（小车移动到极限位置时的支反力 N_z ）、弯矩（小车运行惯性引起 M_y ，其作用面高于螺栓连接面；偏轨布置的引起 M_x ），其中弯矩会引起外排螺栓受拉。单根端梁上的一组螺栓群布置见图 5.1。

The internal force acting on the bolt group is shear press (caused by the travelling inertia of hoist trolley and trolley Q_x , Q_y , M_z), pressure (bearing reaction N_z when the hoist trolley moving to the limit position) and bending moment (caused by the travelling inertia of hoist trolley M_y , its acting surface is higher than the bolt connecting surface; M_x caused by the deflection layout). Wherein, the bending moment will lead to outer-row bolts being pulled. The layout of one bolt group on single end beam is shown in Fig. 5.1.

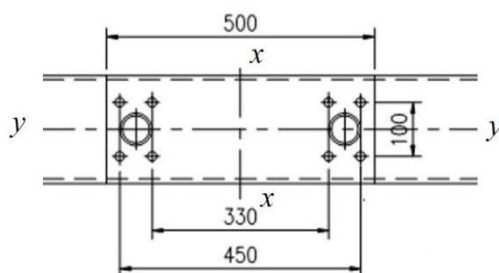


图 5.1 螺栓群的布置图

Fig. 5.1 Layout of Bolt Group

外排螺栓布置间距不能超过 $8d$ 或 $12\delta_{\min}$ 中的最小值，即 $12 \times 12 = 144\text{mm}$ ；最小间距不能小于 $3d = 3 \times 12 = 36\text{mm}$ ，满足布置要求。

The distance between outer row of bolts cannot exceed the minimum value in $8d$ or $12\delta_{\min}$, $12 \times 12 = 144\text{mm}$; the minimum distance shall not be less than $3d = 3 \times 12 = 36\text{mm}$, meeting the layout requirements.

螺栓群的惯性矩

Inertia Moment of Bolt Group

$$I_y = \sum n_i x_i^2 = 4 \times (50^2 + 50^2) = 0.2 \times 10^5 \text{ mm}^2$$

$$I_x = \sum n_i y_i^2 = 4 \times (165^2 + 225^2) = 3.114 \times 10^5 \text{ mm}^2$$

小车运行轨道顶面到螺栓接触面的高度 $h = 268\text{mm}$

Height between the top surface of the hoist trolley rail and the contact surface of the bolt: $h = 268\text{mm}$

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职务: 机械工程师

Position: Mechanical Engineer



剪力 $Q_x = \phi_5 LATT = 1.5 \times 14.62 / 4 = 5.48 \text{ kN}$

Shearing press $Q_x = \phi_5 LATT = 1.5 \times 14.62 / 4 = 5.48 \text{ kN}$

剪力 $Q_y = \phi_5 LATG = 1.5 \times 20.99 / 4 = 7.87 \text{ kN}$

Shearing press $Q_y = \phi_5 LATG = 1.5 \times 20.99 / 4 = 7.87 \text{ kN}$

剪力的合力 $Q = \sqrt{Q_x^2 + Q_y^2} = 9.60 \text{ kN}$

Resulting shear press $Q = \sqrt{Q_x^2 + Q_y^2} = 9.60 \text{ kN}$

压力 $N_z = R_{\max} / 2 = 95.0 / 2 = 47.5 \text{ kN}$

Pressure $N_z = R_{\max} / 2 = 95.0 / 2 = 47.5 \text{ kN}$

弯矩 $M_y = Q_x h = 9.60 \times 0.268 = 2.57 \text{ kN} \cdot \text{m}$ (小车惯性引起)

Bending moment $M_y = Q_x h = 9.60 \times 0.268 = 2.57 \text{ kN} \cdot \text{m}$ (caused by hoist trolley inertia)

弯矩 $M_x = R_{\max} b = 47.5 \times 0.268 = 12.73 \text{ kN} \cdot \text{m}$ (轮压由于偏轨引起)

Bending moment $M_x = R_{\max} b = 47.5 \times 0.268 = 12.73 \text{ kN} \cdot \text{m}$ (the wheel load is caused by deflection)

对于最外侧的螺栓 ($x_1 = 50 \text{ mm}$, $y_1 = 225 \text{ mm}$), 弯矩引起的拉力为

For the outermost bolts ($x_1 = 50 \text{ mm}$, $y_1 = 225 \text{ mm}$), the tensile force caused by bending moment is:

$$N_{zt}^1 = \frac{M_y}{I_y} x_1 + \frac{M_x}{I_x} y_1$$

$$= 2.57 \times 10^6 / (0.2 \times 10^5 / 0.05) + 12.73 \times 10^6 / (3.114 \times 10^5 / 0.225) = 15.62 \text{ kN}$$

单个螺栓的轴向合力为

Resultant axial force of single bolt:

$$P_{t1} = N_{zt}^1 - N_z / 8 = 15.62 - 47.5 / 8 = 9.69 \text{ kN}$$

单个螺栓所承受的剪力为 $P_v = \frac{1}{n} \sqrt{Q_x^2 + Q_y^2} = 1.20 \text{ kN}$

Shearing force on single bolt: $P_v = \frac{1}{n} \sqrt{Q_x^2 + Q_y^2} = 1.20 \text{ kN}$

5.2 螺栓的许用承载能力

5.2 Bolt Allowable Bearing Capacity

采用 M16 螺栓, 螺栓的有效面积 $A_t = 150.35 \text{ mm}^2$ 。

M16 bolt is adopted, and the effective area of the bolt is $A_t = 150.35 \text{ mm}^2$.

根据 DIN 15018 表 12, $[\tau'] = 168 \text{ MPa}$, $[\sigma'] = 140 \text{ MPa}$ (5.6 级), 则

According to Table 12 in DIN 15018, $[\tau'] = 168 \text{ MPa}$, $[\sigma'] = 140 \text{ MPa}$ (Grade 5.6)

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单个螺栓的许用抗剪能力 $[P_v] = A_t[\tau^l] = 150.35 \times 168 \times 10^{-3} = 25.26 \text{ kN}$

Allowable shearing resistance of single bolt: $[P_v] = A_t[\tau^l] = 150.35 \times 168 \times 10^{-3} = 25.26 \text{ kN}$

单个螺栓的许用抗拉能力 $[P_t] = A_t[\sigma^l] = 150.35 \times 140 \times 10^{-3} = 20.05 \text{ kN}$

Allowable tensile capacity of single bolt: $[P_t] = A_t[\sigma^l] = 150.35 \times 140 \times 10^{-3} = 20.05 \text{ kN}$

被连接板的材料为 Q355B，因此螺栓孔壁的承压许用应力 $[\sigma_c^l] = 1.4[\sigma] = 1.4 \times 218 = 305.2 \text{ MPa}$ ，
则孔壁许用挤压能力

The connecting plate is made of Q355B, so the allowable pressure stress of the bolt hole wall is:
 $[\sigma_c^l] = 1.4[\sigma] = 1.4 \times 218 = 305.2 \text{ MPa}$, the allowable extrusion capacity of the hole wall is:

$$[P_c] = d \Sigma \delta [\sigma_c^l] = 12 \times 18 \times 305.2 \times 10^{-3} = 65.92 \text{ kN}$$

5.3 螺栓连接校核

5.3 Check of Bolt Connection

校核公式为 $\sqrt{\left(\frac{P_t}{[P_t]}\right)^2 + \left(\frac{P_v}{[P_v]}\right)^2} \leq 1$

The check formula is as follows: $\sqrt{\left(\frac{P_t}{[P_t]}\right)^2 + \left(\frac{P_v}{[P_v]}\right)^2} \leq 1$

螺栓 1: $\sqrt{\left(\frac{9.69}{25.26}\right)^2 + \left(\frac{1.20}{20.05}\right)^2} = 0.39 < 1$ 满足要求!

Bolt 1: $\sqrt{\left(\frac{9.69}{25.26}\right)^2 + \left(\frac{1.20}{20.05}\right)^2} = 0.39 < 1$, it meets the requirements!

$$P_v \leq \frac{[P_c]}{1.2} = 65.92 / 1.2 = 54.93 \text{ kN} \text{ 满足要求!}$$

$$P_v \leq \frac{[P_c]}{1.2} = 234.39 / 1.2 = 195.3 \text{ MPa, it meets the requirements!}$$

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